8 The stem and leaf diagram shows information about the heights, in cm, of the boys in a class.

14	0	2	9			
15	1	1	3	5	7	
16	2	4	5	7	8	9
17	6	6	7	9		
18	0	0	1			

Key: 15 | 1 represents 151 cm

(a) Find the median height.

21 boys: middle position = (21 + 1) ÷ 2 = 11.

The 11th height is 165 cm.

1 mark
Median 165 cm

..... cm

(1)

The girls in the class have a median height of 162 cm.
Their heights have a range of 45 cm.

(b) Compare the distribution of the heights of the boys with the distribution of the heights of the girls.

Boys have the higher median: 165 cm > 162 cm,

so the boys are generally taller.

Boys' range = 181 – 140 = 41 cm.

41 cm < 45 cm: boys' heights are less spread out.

1 mark
Median compared in context

1 mark
Range 41 cm; smaller spread

(2)

(Total for Question 8 is 3 marks)